ELECTRON DELOCALIZATION

Tome 1

Reduced Densities and Their Units

One-electron density, pair density and exchange-correlation density

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Scope and Notation

This tome defines the one-electron density, the pair density and the exchange-correlation density of an N-electron system. For each quantity, it states the definition, the normalization, the units, the spin-resolved forms and the orbital representation. Tome 2 uses these quantities to define atomic populations, the localization index and the delocalization index.

Prerequisites. Quantum mechanics of many-electron systems, Slater determinants, Hartree–Fock theory, and integral calculus in three dimensions.

Conventions. All quantities are expressed in atomic units (Section 1.3) unless a different unit is stated. The wave function Ψ is normalized to one. Spin-summed quantities depend on positions only and carry the argument $\mathbf{r}$; spin-dependent quantities carry the argument $\mathbf{x} = (\mathbf{r}, \sigma)$.

Symbol	Quantity	Unit (a.u.)
N	number of electrons	dimensionless
$\mathbf{r}$	position vector	a_0
σ	spin variable, $\sigma \in \{\uparrow, \downarrow\}$	dimensionless
$\mathbf{x} = (\mathbf{r}, \sigma)$	space-spin coordinate	—
Ψ	N-electron wave function	$a_0^{-3N/2}$
ϕ_i	spatial orbital	$a_0^{-3/2}$
$\rho(\mathbf{r})$	one-electron density	a_0^{-3}
γ_1	one-electron reduced density matrix	a_0^{-3}
ρ_2	pair density	a_0^{-6}
γ_{xc}	exchange-correlation density	a_0^{-6}
h_{xc}	exchange-correlation hole	a_0^{-3}

1. Foundations

1.1 Space-spin coordinates

Each electron is described by a space-spin coordinate $\mathbf{x} = (\mathbf{r}, \sigma)$, where $\mathbf{r} \in \mathbb{R}^3$ is the position and $\sigma \in \{\uparrow, \downarrow\}$ is the spin variable. Integration over $\mathbf{x}$ is defined as

$$\int \mathrm{d}\mathbf{x}\, f(\mathbf{x}) = \sum_{\sigma} \int \mathrm{d}\mathbf{r}\, f(\mathbf{r}, \sigma) = \int \mathrm{d}\mathbf{r}\, [f(\mathbf{r}, \uparrow) + f(\mathbf{r}, \downarrow)]. \tag{1.1}$$

The spin sum is dimensionless. The element $\mathrm{d}\mathbf{x}$ therefore has the dimension of a volume, $[L^3]$.

1.2 The N-electron wave function

The wave function $\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N)$ of N electrons is antisymmetric under the exchange of the coordinates of any two electrons,

$$\Psi(\dots, \mathbf{x}_i, \dots, \mathbf{x}_j, \dots) = -\Psi(\dots, \mathbf{x}_j, \dots, \mathbf{x}_i, \dots), \tag{1.2}$$

and normalized,

$$\int |\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N)|^2 \, \mathrm{d}\mathbf{x}_1 \cdots \mathrm{d}\mathbf{x}_N = 1. \tag{1.3}$$

The quantity $|\Psi|^2 \, \mathrm{d}\mathbf{x}_1 \cdots \mathrm{d}\mathbf{x}_N$ is the probability of finding electron 1 in $\mathrm{d}\mathbf{x}_1$, electron 2 in $\mathrm{d}\mathbf{x}_2$, and so on. The element $\mathrm{d}\mathbf{x}_1 \cdots \mathrm{d}\mathbf{x}_N$ has dimension $[L^{3N}]$. Hence $|\Psi|^2$ has dimension $[L^{-3N}]$ and Ψ has dimension $\left[L^{-3N/2}\right]$.

Antisymmetry implies that $|\Psi|^2$ is invariant under any permutation of the electron labels. The electrons are indistinguishable, and every reduced quantity defined below is the same for each choice of the retained electrons. This property produces the prefactors N and $N(N-1)$ in the definitions of the density and of the pair density.

1.3 Atomic units

Atomic units are defined by $\hbar = m_e = e = 4\pi\varepsilon_0 = 1$. The base units and their SI values are listed in Table 1.1.

Quantity	Atomic unit	Symbol	SI value
Length	Bohr radius	a_0	$5.29177210903 \times 10^{-11}$ m
Energy	Hartree	E_h	$4.3597447222 \times 10^{-18}$ J $= 27.211386$ eV
Charge	elementary charge	e	$1.602176634 \times 10^{-19}$ C
Mass	electron mass	m_e	$9.1093837015 \times 10^{-31}$ kg
Dipole moment	ea_0	—	$8.4783536 \times 10^{-30}$ C m $= 2.541746$ D

Table 1.1: Atomic units (CODATA 2018 values).

The conversion $1\,a_0 = 0.529177210903$ Å applies to all length-dependent quantities. In atomic units, the Coulomb interaction of two unit charges at distance r is $1/r$, with $1/r$ expressed in units of E_h when r is expressed in units of a_0.

2. The One-Electron Density

2.1 Definition

Definition 2.1 — One-electron density

The spin-dependent one-electron density is

$$\rho(\mathbf{x}_1) = N \int |\Psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N)|^2 \, \mathrm{d}\mathbf{x}_2 \cdots \mathrm{d}\mathbf{x}_N. \tag{2.1}$$

The spin-resolved densities are $\rho^{\uparrow}(\mathbf{r}) = \rho(\mathbf{r}, \uparrow)$ and $\rho^{\downarrow}(\mathbf{r}) = \rho(\mathbf{r}, \downarrow)$. The total density and the spin density are

$$\rho(\mathbf{r}) = \sum_{\sigma} \rho(\mathbf{r}, \sigma) = \rho^{\uparrow}(\mathbf{r}) + \rho^{\downarrow}(\mathbf{r}), \qquad \rho_s(\mathbf{r}) = \rho^{\uparrow}(\mathbf{r}) - \rho^{\downarrow}(\mathbf{r}). \tag{2.2}$$

The product $\rho(\mathbf{r})\,\mathrm{d}\mathbf{r}$ is the expected number of electrons in the volume element $\mathrm{d}\mathbf{r}$ at $\mathbf{r}$. The factor N in Eq. (2.1) accounts for the fact that any of the N identical electrons can be the electron found in $\mathrm{d}\mathbf{r}$; the N events are mutually exclusive, and their probabilities add.

2.2 Normalization

Integration of Eq. (2.1) over $\mathbf{x}_1$ and use of the normalization of Ψ give

$$\int \rho(\mathbf{r})\,\mathrm{d}\mathbf{r} = N, \qquad \int \rho^{\sigma}(\mathbf{r})\,\mathrm{d}\mathbf{r} = N^{\sigma}, \qquad N^{\uparrow} + N^{\downarrow} = N, \tag{2.3}$$

where N^{σ} is the number of electrons with spin σ. The integral of the spin density is $\int \rho_s \,\mathrm{d}\mathbf{r} = N^{\uparrow} - N^{\downarrow}$. The function $\rho(\mathbf{r})/N$ is a probability density that integrates to one.

2.3 Units

The density $\rho(\mathbf{r})$ is a number density: the integral over a volume yields a number of electrons, which is dimensionless. The dimension of ρ is therefore the inverse of a volume,

$$[\rho] = [L^{-3}] \quad \Rightarrow \quad \text{unit in a.u.: } a_0^{-3} \text{ (electrons per bohr}^3\text{)}. \tag{2.4}$$

The electronic charge density is $\rho_e(\mathbf{r}) = -e\,\rho(\mathbf{r})$, with the unit $e\,a_0^{-3}$. Table 2.1 lists the conversion factors.

Quantity	Value of 1 a.u.	Unit
Number density ρ	6.7483	Å^{-3}
Number density ρ	6.7483×10^{30}	m^{-3}
Charge density $-e\rho$	1.0812×10^{12}	$\text{C}\,\text{m}^{-3}$

Table 2.1: Conversion of the one-electron density from atomic units.

2.4 Properties

Property 2.1 — Analytical properties of ρ

1. Non-negativity: $\rho(\mathbf{r}) \geq 0$ for all $\mathbf{r}$.
2. Cusp condition (Kato): at a nucleus A of charge Z_A located at $\mathbf{R}_A$, the spherical average $\bar{\rho}$ of ρ about $\mathbf{R}_A$ satisfies
$$\left.\frac{\partial \bar{\rho}}{\partial r}\right|_{r=0} = -2Z_A\,\rho(\mathbf{R}_A) \quad \text{(a.u.)}. \tag{2.5}$$
3. Asymptotic decay: for large $|\mathbf{r}|$, $\rho(\mathbf{r}) \propto \exp\!\left(-2\sqrt{2I}\,|\mathbf{r}|\right)$, where I is the first ionization energy in E_h.

2.5 Orbital representation

For a single determinant of spin orbitals $\chi_i(\mathbf{x}) = \phi_i(\mathbf{r})\eta_i(\sigma)$, the density is

$$\rho(\mathbf{r}) = \sum_{i=1}^{N} |\phi_i(\mathbf{r})|^2. \tag{2.6}$$

For a closed-shell determinant with $N/2$ doubly occupied spatial orbitals, $\rho(\mathbf{r}) = 2\sum_{i=1}^{N/2} |\phi_i(\mathbf{r})|^2$. Each orbital is normalized, $\int |\phi_i|^2 \mathrm{d}\mathbf{r} = 1$, so ϕ_i has the unit $a_0^{-3/2}$.

In a basis $\{\chi_\mu\}$ with overlap matrix $S_{\mu\nu} = \int \chi_\mu^* \chi_\nu \,\mathrm{d}\mathbf{r}$ and density matrix $P_{\mu\nu} = \sum_i n_i C_{\mu i} C_{\nu i}^*$, where n_i is the occupation number of orbital i,

$$\rho(\mathbf{r}) = \sum_{\mu\nu} P_{\mu\nu}\,\chi_\mu(\mathbf{r})\chi_\nu^*(\mathbf{r}), \qquad \int \rho\,\mathrm{d}\mathbf{r} = \mathrm{Tr}(\mathbf{PS}) = N. \tag{2.7}$$

2.6 Expectation values of one-electron operators

For an operator $\hat{F} = \sum_{i=1}^{N} f(\mathbf{r}_i)$,

$$\langle \hat{F} \rangle = \int f(\mathbf{r})\,\rho(\mathbf{r})\,\mathrm{d}\mathbf{r}. \tag{2.8}$$

Two examples are the electron–nucleus attraction energy and the electronic contribution to the dipole moment,

$$V_{ne} = -\sum_A Z_A \int \frac{\rho(\mathbf{r})}{|\mathbf{r} - \mathbf{R}_A|}\,\mathrm{d}\mathbf{r} \quad [E_\text{h}], \qquad \mu = \sum_A Z_A \mathbf{R}_A - \int \mathbf{r}\,\rho(\mathbf{r})\,\mathrm{d}\mathbf{r} \quad [e\,a_0]. \tag{2.9}$$

The unit of V_{ne} follows from $a_0^{-3}\cdot a_0^{-1}\cdot a_0^{3} = a_0^{-1}$ multiplied by the Coulomb constant $e^2/4\pi\varepsilon_0 = E_\mathrm{h}\,a_0$.

Example 2.1 — Hydrogen atom, ground state

The orbital is $\phi_{1s}(r) = (\pi a_0^3)^{-1/2} e^{-r/a_0}$ and the density is

$$\rho(r) = \frac{1}{\pi a_0^3}\, e^{-2r/a_0}. \tag{2.10}$$

Normalization: $\int \rho\,\mathrm{d}\mathbf{r} = \dfrac{4}{a_0^3}\int_0^\infty r^2 e^{-2r/a_0}\,\mathrm{d}r = \dfrac{4}{a_0^3}\cdot\dfrac{2a_0^3}{8} = 1$.

Value at the nucleus: $\rho(0) = 1/(\pi a_0^3) = 0.31831\ a_0^{-3} = 2.1480\ \text{Å}^{-3}$.

Cusp: $\mathrm{d}\rho/\mathrm{d}r = -(2/a_0)\rho$, hence $\mathrm{d}\rho/\mathrm{d}r|_{r=0} = -2Z\rho(0)$ with $Z = 1$ in atomic units.

Population in a sphere of radius R, with $x = R/a_0$:

$$N(R) = \int_0^R \rho\, 4\pi r^2 \mathrm{d}r = 1 - e^{-2x}\,(1 + 2x + 2x^2). \tag{2.11}$$

3. The Pair Density

3.1 Definition

Definition 3.1 — Pair density

The spin-dependent pair density is

$$\rho_2(\mathbf{x}_1, \mathbf{x}_2) = N(N-1) \int |\Psi(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_N)|^2 \, \mathrm{d}\mathbf{x}_3 \cdots \mathrm{d}\mathbf{x}_N. \tag{3.1}$$

The product $\rho_2(\mathbf{x}_1, \mathbf{x}_2)\, \mathrm{d}\mathbf{x}_1 \mathrm{d}\mathbf{x}_2$ is the expected number of ordered pairs of distinct electrons in which one electron is in $\mathrm{d}\mathbf{x}_1$ and the other electron is in $\mathrm{d}\mathbf{x}_2$. The factor $N(N-1)$ is the number of ordered pairs of distinct electrons. The pair density is symmetric,

$$\rho_2(\mathbf{x}_1, \mathbf{x}_2) = \rho_2(\mathbf{x}_2, \mathbf{x}_1), \tag{3.2}$$

and normalized to the number of ordered pairs,

$$\iint \rho_2(\mathbf{x}_1, \mathbf{x}_2)\, \mathrm{d}\mathbf{x}_1 \mathrm{d}\mathbf{x}_2 = N(N-1). \tag{3.3}$$

Some authors normalize the pair density to $N(N-1)/2$, the number of unordered pairs. The normalization $N(N-1)$ is used throughout this work.

Units. The function $|\Psi|^2$ has dimension $[L^{-3N}]$, and the integration over $N-2$ coordinates contributes $[L^{3(N-2)}]$. Hence $[\rho_2] = [L^{-6}]$, with the unit a_0^{-6} in atomic units.

3.2 Spin-resolved and spin-summed forms

The spin-summed pair density depends on two positions,

$$\rho_2(\mathbf{r}_1, \mathbf{r}_2) = \sum_{\sigma_1 \sigma_2} \rho_2(\mathbf{r}_1 \sigma_1, \mathbf{r}_2 \sigma_2) = \rho_2^{\uparrow\uparrow} + \rho_2^{\uparrow\downarrow} + \rho_2^{\downarrow\uparrow} + \rho_2^{\downarrow\downarrow}, \tag{3.4}$$

with the spin components $\rho_2^{\sigma\sigma'}(\mathbf{r}_1, \mathbf{r}_2) = \rho_2(\mathbf{r}_1\sigma, \mathbf{r}_2\sigma')$. The normalization of each component is

$$\iint \rho_2^{\sigma\sigma'}(\mathbf{r}_1, \mathbf{r}_2)\, \mathrm{d}\mathbf{r}_1 \mathrm{d}\mathbf{r}_2 = N^{\sigma}(N^{\sigma'} - \delta_{\sigma\sigma'}). \tag{3.5}$$

The sum over the four components gives $(N^{\uparrow} + N^{\downarrow})^2 - (N^{\uparrow} + N^{\downarrow}) = N(N-1)$. The component relation $\rho_2^{\uparrow\downarrow}(\mathbf{r}_1, \mathbf{r}_2) = \rho_2^{\downarrow\uparrow}(\mathbf{r}_2, \mathbf{r}_1)$ follows from the symmetry of ρ_2.

3.3 Reduction to the one-electron density

Integration of Eq. (3.1) over $\mathbf{x}_2$ gives

$$\int \rho_2(\mathbf{x}_1, \mathbf{x}_2)\, \mathrm{d}\mathbf{x}_2 = N(N-1) \int |\Psi|^2 \mathrm{d}\mathbf{x}_2 \cdots \mathrm{d}\mathbf{x}_N = (N-1)\, \rho(\mathbf{x}_1), \tag{3.6}$$

and the spin-summed form

$$\int \rho_2(\mathbf{r}_1, \mathbf{r}_2)\, \mathrm{d}\mathbf{r}_2 = (N-1)\, \rho(\mathbf{r}_1). \tag{3.7}$$

The units agree: $a_0^{-6} \cdot a_0^3 = a_0^{-3}$.

3.4 Conditional density

For $\rho(\mathbf{r}_1) > 0$, the conditional density is

$$\rho(\mathbf{r}_2|\mathbf{r}_1) = \frac{\rho_2(\mathbf{r}_1, \mathbf{r}_2)}{\rho(\mathbf{r}_1)}, \qquad \int \rho(\mathbf{r}_2|\mathbf{r}_1)\, \mathrm{d}\mathbf{r}_2 = N - 1. \tag{3.8}$$

The function $\rho(\mathbf{r}_2|\mathbf{r}_1)$ is the density of the remaining $N-1$ electrons at $\mathbf{r}_2$ given that one electron is located at $\mathbf{r}_1$. Its unit is a_0^{-3}.

3.5 Independent-electron reference

For uncorrelated, distinguishable electrons the pair density is the product $\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)$. This product integrates to N^2, whereas ρ_2 integrates to $N(N-1)$:

$$\iint [\rho_2(\mathbf{r}_1, \mathbf{r}_2) - \rho(\mathbf{r}_1)\rho(\mathbf{r}_2)]\mathrm{d}\mathbf{r}_1\mathrm{d}\mathbf{r}_2 = -N. \tag{3.9}$$

The deficit $-N$ corresponds to the exclusion of one electron from the neighborhood of each reference electron. Chapter 4 defines this difference as the exchange-correlation density.

3.6 Pauli principle

The wave function Ψ vanishes when two electrons have identical space-spin coordinates, because of antisymmetry. Therefore

$$\rho_2(\mathbf{x}, \mathbf{x}) = 0, \qquad \rho_2^{\sigma\sigma}(\mathbf{r}, \mathbf{r}) = 0. \tag{3.10}$$

Two electrons with the same spin have zero probability of occupying the same point.

3.7 Electron repulsion energy

The electron–electron repulsion energy is a functional of the pair density only,

$$V_{ee} = \frac{1}{2} \iint \frac{\rho_2(\mathbf{r}_1, \mathbf{r}_2)}{r_{12}}\, \mathrm{d}\mathbf{r}_1\mathrm{d}\mathbf{r}_2, \qquad r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|. \tag{3.11}$$

Dimensional analysis: $a_0^{-6} \cdot a_0^{-1} \cdot a_0^6 = a_0^{-1}$, multiplied by $e^2/4\pi\varepsilon_0 = E_\mathrm{h}\, a_0$, gives E_h. The factor $1/2$ corrects for the double counting of ordered pairs.

3.8 Single-determinant pair density

Definition 3.2 — One-electron reduced density matrix

$$\gamma_1(\mathbf{x}_1;\mathbf{x}_1') = N\int \Psi(\mathbf{x}_1,\mathbf{x}_2,\dots,\mathbf{x}_N)\,\Psi^*(\mathbf{x}_1',\mathbf{x}_2,\dots,\mathbf{x}_N)\,\mathrm{d}\mathbf{x}_2\cdots\mathrm{d}\mathbf{x}_N. \tag{3.12}$$

Its diagonal is the density, $\gamma_1(\mathbf{x};\mathbf{x}) = \rho(\mathbf{x})$, and its unit is a_0^{-3}.

For a single determinant of spin orbitals χ_i, $\gamma_1(\mathbf{x}_1;\mathbf{x}_2) = \sum_i \chi_i(\mathbf{x}_1)\chi_i^*(\mathbf{x}_2)$ and

$$\rho_2(\mathbf{x}_1,\mathbf{x}_2) = \rho(\mathbf{x}_1)\rho(\mathbf{x}_2) - |\gamma_1(\mathbf{x}_1;\mathbf{x}_2)|^2. \tag{3.13}$$

The spin-summed form is $\rho_2(\mathbf{r}_1,\mathbf{r}_2) = \rho(\mathbf{r}_1)\rho(\mathbf{r}_2) - \sum_\sigma |\gamma^\sigma(\mathbf{r}_1,\mathbf{r}_2)|^2$, where γ^σ is the spin-σ block of γ_1. For a closed-shell determinant, $\gamma^\uparrow = \gamma^\downarrow = \gamma/2$ with $\gamma(\mathbf{r}_1,\mathbf{r}_2) = 2\sum_{i=1}^{N/2}\phi_i(\mathbf{r}_1)\phi_i^*(\mathbf{r}_2)$, hence

$$\rho_2(\mathbf{r}_1,\mathbf{r}_2) = \rho(\mathbf{r}_1)\rho(\mathbf{r}_2) - \tfrac{1}{2}|\gamma(\mathbf{r}_1,\mathbf{r}_2)|^2. \tag{3.14}$$

Example 3.1 — Two-electron singlet in a single determinant

Both electrons occupy the spatial orbital ϕ, so $\rho = 2|\phi|^2$ and $\gamma(\mathbf{r}_1,\mathbf{r}_2) = 2\phi(\mathbf{r}_1)\phi^*(\mathbf{r}_2)$. Equation (3.14) gives

$$\rho_2(\mathbf{r}_1,\mathbf{r}_2) = 4|\phi_1|^2|\phi_2|^2 - 2|\phi_1|^2|\phi_2|^2 = 2|\phi_1|^2|\phi_2|^2 = \tfrac{1}{2}\rho(\mathbf{r}_1)\rho(\mathbf{r}_2). \tag{3.15}$$

Normalization: $\iint \rho_2 = \tfrac{1}{2}N^2 = 2 = N(N-1)$ for $N = 2$.

4. The Exchange-Correlation Density and Hole

4.1 Exchange-correlation density

Definition 4.1 — Exchange-correlation density

$$\gamma_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2) = \rho_2(\mathbf{r}_1,\mathbf{r}_2) - \rho(\mathbf{r}_1)\rho(\mathbf{r}_2). \tag{4.1}$$

The function γ_{xc} is symmetric in its arguments and has the unit a_0^{-6}. It measures the deviation of the pair density from the product of one-electron densities. Integration over $\mathbf{r}_2$ with Eq. (3.7) and $\int \rho(\mathbf{r}_2)\mathrm{d}\mathbf{r}_2 = N$ gives

$$\int \gamma_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)\,\mathrm{d}\mathbf{r}_2 = (N-1)\rho(\mathbf{r}_1) - N\rho(\mathbf{r}_1) = -\rho(\mathbf{r}_1), \qquad \iint \gamma_{\mathrm{xc}}\,\mathrm{d}\mathbf{r}_1\mathrm{d}\mathbf{r}_2 = -N. \tag{4.2}$$

4.2 Exchange-correlation hole

Definition 4.2 — Exchange-correlation hole

For $\rho(\mathbf{r}_1) > 0$,

$$h_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2) = \frac{\gamma_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)}{\rho(\mathbf{r}_1)}, \qquad \rho_2(\mathbf{r}_1,\mathbf{r}_2) = \rho(\mathbf{r}_1)[\rho(\mathbf{r}_2) + h_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)]. \tag{4.3}$$

The hole $h_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)$ is the change in the density at $\mathbf{r}_2$ caused by the presence of an electron at $\mathbf{r}_1$. Its unit is a_0^{-3}. Comparison with the conditional density gives $\rho(\mathbf{r}_2|\mathbf{r}_1) = \rho(\mathbf{r}_2) + h_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)$.

Property 4.1 — Hole sum rule and lower bound

$$\int h_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)\,\mathrm{d}\mathbf{r}_2 = -1, \qquad h_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2) \geq -\rho(\mathbf{r}_2). \tag{4.4}$$

The sum rule follows from Eq. (4.2). The lower bound follows from $\rho(\mathbf{r}_2|\mathbf{r}_1) \geq 0$.

The sum rule states that the hole removes exactly one electron from the surroundings of the reference electron. This removal compensates the inclusion of the reference electron in the product $\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)$, which treats every electron as interacting with the full density, including its own contribution.

4.3 Spin structure and the Pauli condition

The pair density has four spin components, and the hole is defined for each pair of spins. For a reference electron of spin σ at $\mathbf{r}_1$ and a second electron of spin σ' at $\mathbf{r}_2$,

$$h_{\mathrm{xc}}^{\sigma\sigma'}(\mathbf{r}_1,\mathbf{r}_2) = \frac{\rho_2^{\sigma\sigma'}(\mathbf{r}_1,\mathbf{r}_2)}{\rho^{\sigma}(\mathbf{r}_1)} - \rho^{\sigma'}(\mathbf{r}_2). \tag{4.5}$$

The condition $\rho_2^{\sigma\sigma}(\mathbf{r},\mathbf{r}) = 0$ yields the exact result

$$h_{\mathrm{xc}}^{\sigma\sigma}(\mathbf{r},\mathbf{r}) = -\rho^{\sigma}(\mathbf{r}). \tag{4.6}$$

At coalescence, the same-spin hole cancels the density of the same-spin electrons completely. The opposite-spin hole $h_{\mathrm{xc}}^{\sigma\bar{\sigma}}(\mathbf{r},\mathbf{r})$ has no such constraint.

4.4 Exchange and correlation components

For a single determinant, the pair density is given by Eq. (3.13). The corresponding hole is the exchange (Fermi) hole,

$$h_x^{\sigma\sigma}(\mathbf{r}_1,\mathbf{r}_2) = -\frac{|\gamma^{\sigma}(\mathbf{r}_1,\mathbf{r}_2)|^2}{\rho^{\sigma}(\mathbf{r}_1)}, \qquad h_x^{\sigma\bar{\sigma}} = 0, \tag{4.7}$$

and $\int h_x^{\sigma\sigma}(\mathbf{r}_1,\mathbf{r}_2)\mathrm{d}\mathbf{r}_2 = -1$, because $\int |\gamma^{\sigma}(\mathbf{r}_1,\mathbf{r}_2)|^2 \mathrm{d}\mathbf{r}_2 = \rho^{\sigma}(\mathbf{r}_1)$ for an idempotent γ^{σ}. The exchange hole originates from antisymmetry only and acts between electrons of the same spin.

For a correlated wave function, the correlation hole is the difference between the hole of Ψ and the exchange hole of a reference determinant,

$$h_c = h_{\mathrm{xc}} - h_x, \qquad \int h_c(\mathbf{r}_1,\mathbf{r}_2)\,\mathrm{d}\mathbf{r}_2 = 0. \tag{4.8}$$

The correlation hole redistributes the density of the hole without changing its integral. It results from the Coulomb repulsion between electrons of either spin.

4.5 Electron repulsion energy

Insertion of Eq. (4.3) in V_{ee} separates the classical Coulomb energy from the exchange-correlation energy,

$$V_{ee} = J + E_{xc}, \qquad J = \frac{1}{2}\iint \frac{\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)}{r_{12}}\mathrm{d}\mathbf{r}_1\mathrm{d}\mathbf{r}_2, \qquad E_{xc} = \frac{1}{2}\iint \frac{\gamma_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)}{r_{12}}\mathrm{d}\mathbf{r}_1\mathrm{d}\mathbf{r}_2. \tag{4.9}$$

Both terms have the unit E_{h}. The energy E_{xc} is the Coulomb interaction energy of each electron with its exchange-correlation hole.

Example 4.1 — Exchange-correlation quantities of a two-electron singlet

For the determinant of the previous chapter, $\rho_2 = \frac{1}{2}\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)$, hence

$$\gamma_{\mathrm{xc}} = -\tfrac{1}{2}\rho(\mathbf{r}_1)\rho(\mathbf{r}_2), \qquad h_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2) = -\tfrac{1}{2}\rho(\mathbf{r}_2). \tag{4.10}$$

The hole sum rule is satisfied: $\int h_{\mathrm{xc}}\,\mathrm{d}\mathbf{r}_2 = -N/2 = -1$. The exchange-correlation energy is $E_{xc} = -J/2$, so $V_{ee} = J + E_{xc} = J/2$.

5. Units: Summary and Conversion

Table 5.1 summarizes the reduced quantities of this tome. The columns give the dimension of each function, the value of its integral, and the conversion of 1 a.u. to Å and SI units.

Quantity	a.u.	Integral	1 a.u. in Å	1 a.u. in SI
$\rho(\mathbf{r})$	a_0^{-3}	N (over $\mathbf{r}$)	6.7483 Å^{-3}	6.7483×10^{30} m^{-3}
$\gamma_1(\mathbf{r};\mathbf{r}')$	a_0^{-3}	N (trace)	6.7483 Å^{-3}	6.7483×10^{30} m^{-3}
$\rho(\mathbf{r}_2\|\mathbf{r}_1)$	a_0^{-3}	$N-1$ (over $\mathbf{r}_2$)	6.7483 Å^{-3}	6.7483×10^{30} m^{-3}
$h_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)$	a_0^{-3}	-1 (over $\mathbf{r}_2$)	6.7483 Å^{-3}	6.7483×10^{30} m^{-3}
$\rho_2(\mathbf{r}_1,\mathbf{r}_2)$	a_0^{-6}	$N(N-1)$ (over $\mathbf{r}_1,\mathbf{r}_2$)	45.540 Å^{-6}	4.5540×10^{61} m^{-6}
$\gamma_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)$	a_0^{-6}	$-N$ (over $\mathbf{r}_1,\mathbf{r}_2$)	45.540 Å^{-6}	4.5540×10^{61} m^{-6}

Table 5.1: Dimensions, normalizations and unit conversions of the reduced quantities.

Rules for dimensional checks.

1. A function of k positions obtained by integrating $|\Psi|^2$ over the other $N-k$ positions has the dimension $[L^{-3k}]$.
2. Integration over one position multiplies the dimension by $[L^3]$. The integrals of ρ, ρ_2 and γ_{xc} over all positions are dimensionless numbers (N, $N(N-1)$, $-N$).
3. Energies are obtained by multiplying the integrated density function by the Coulomb constant $e^2/4\pi\varepsilon_0 = E_{\mathrm{h}}\, a_0$ and by $1/r_{12}$, of dimension $[L^{-1}]$.

A. Exercises

1. Convert the hydrogen-atom density at the nucleus, $\rho(0) = 1/(\pi a_0^3)$, to Å^{-3}.

2. A system has $N = 10$ electrons. Give $\iint \rho_2 \, \mathrm{d}\mathbf{r}_1 \mathrm{d}\mathbf{r}_2$ and $\iint \gamma_{\mathrm{xc}} \, \mathrm{d}\mathbf{r}_1 \mathrm{d}\mathbf{r}_2$.

3. Convert $\rho = 0.25 \; a_0^{-3}$ to Å^{-3} and to m^{-3}.

4. Compute the population of the hydrogen $1s$ density inside a sphere of radius $R = a_0$.

5. For the two-electron singlet of Chapter 3, show that $E_{xc} = -J/2$.

6. Verify by dimensional analysis that $\frac{1}{2} \iint \rho_2 / r_{12} \, \mathrm{d}\mathbf{r}_1 \mathrm{d}\mathbf{r}_2$ has the dimension of energy.

B. Solutions

1. $\rho(0) = 0.31831\ a_0^{-3}$; with $1\ a_0^{-3} = 6.7483\ \text{Å}^{-3}$, $\rho(0) = 2.148\ \text{Å}^{-3}$.

2. $\iint \rho_2 = N(N-1) = 90$ and $\iint \gamma_{\text{xc}} = -N = -10$.

3. $0.25 \times 6.7483 = 1.687\ \text{Å}^{-3}$, and $0.25 \times 6.7483 \times 10^{30} = 1.687 \times 10^{30}\ \text{m}^{-3}$.

4. With $x = 1$: $N(R) = 1 - e^{-2}(1 + 2 + 2) = 1 - 5e^{-2} = 0.3233$.

5. $E_{xc} = \frac{1}{2} \iint \gamma_{\text{xc}}/r_{12} = -\frac{1}{4} \iint \rho\rho/r_{12} = -\frac{1}{2}J$, because $J = \frac{1}{2} \iint \rho\rho/r_{12}$.

6. $a_0^{-6} \cdot a_0^{-1} \cdot a_0^6 = a_0^{-1}$. Multiplication by $e^2/4\pi\varepsilon_0$, which has the dimension $E_\text{h}\, a_0$, gives E_h.

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